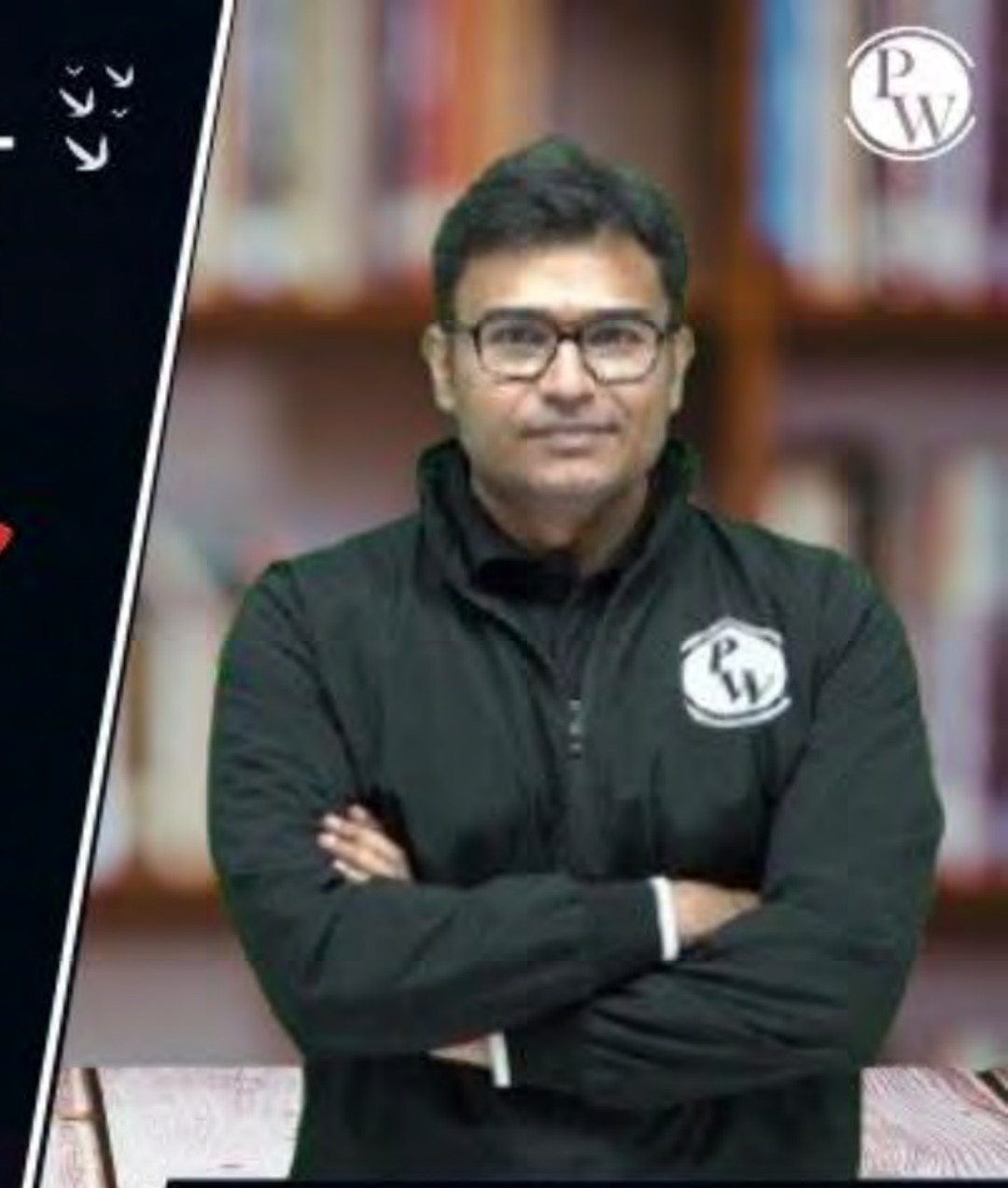
Computer Science & IT

Theory of Computation



Regular languages

Lecture No. 7



By- Ravindra Sir

Recap of Previous Lecture



Topic DFA

Topic NFA

Topic RE

divisibility/

Topics to be Covered

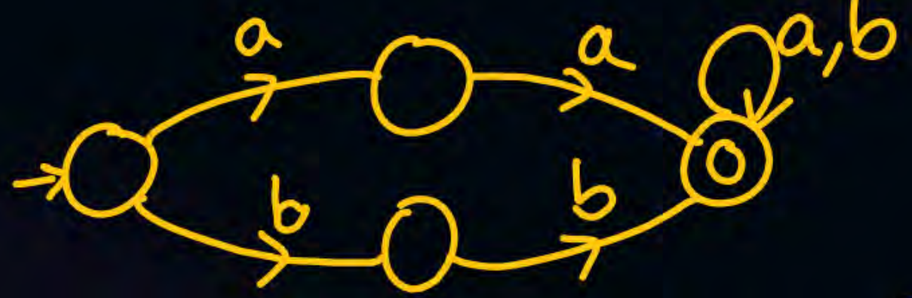


- Topic DFA
- Topic NFA
- Topic RE

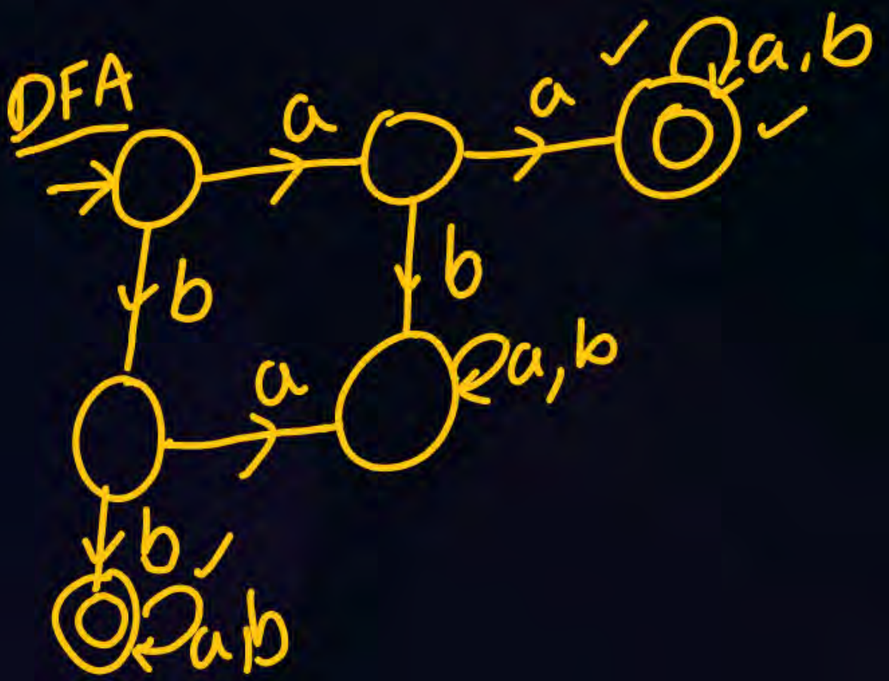
$\Sigma = \{a, b\}$ $L = \{w / \text{first two symbols of } w \text{ are same}\}$

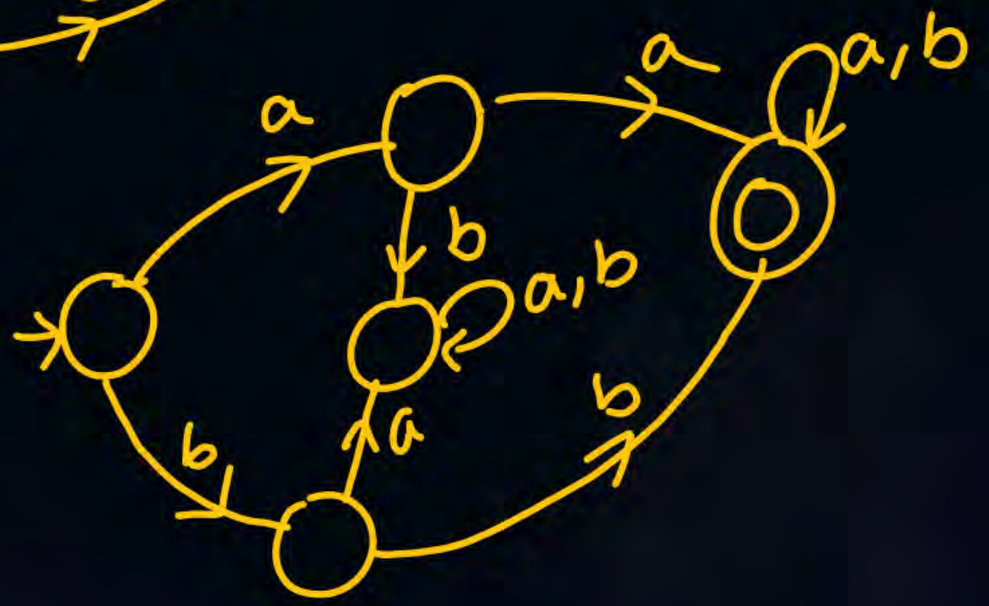
$(aa)(a+b)^* + (bb)(a+b)^*$

aa
ab
ba
bb



DFA

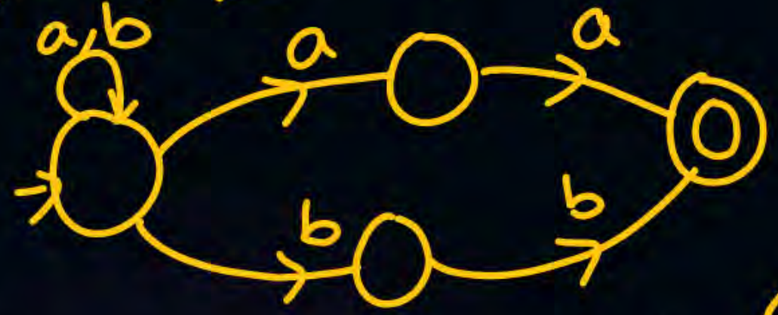


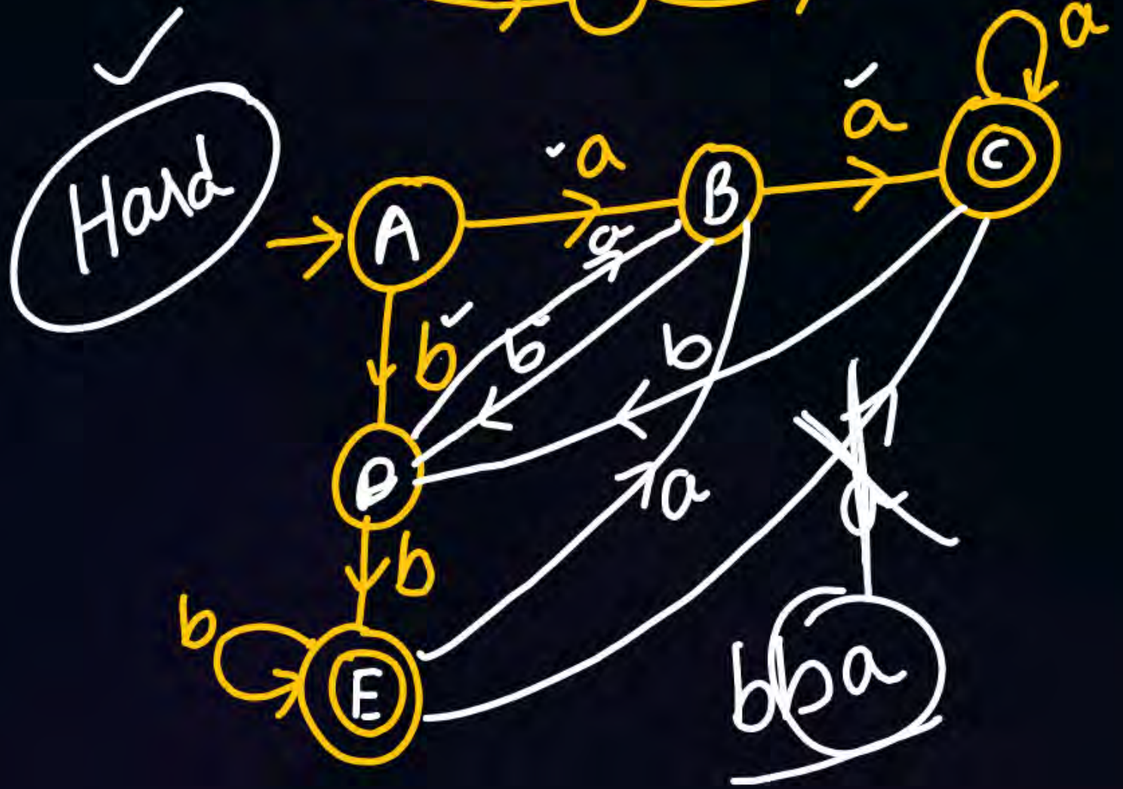


→ last two symbols are same.

$(a+b)^*aa + (a+b)^*bb$

NFA:





D → b@a

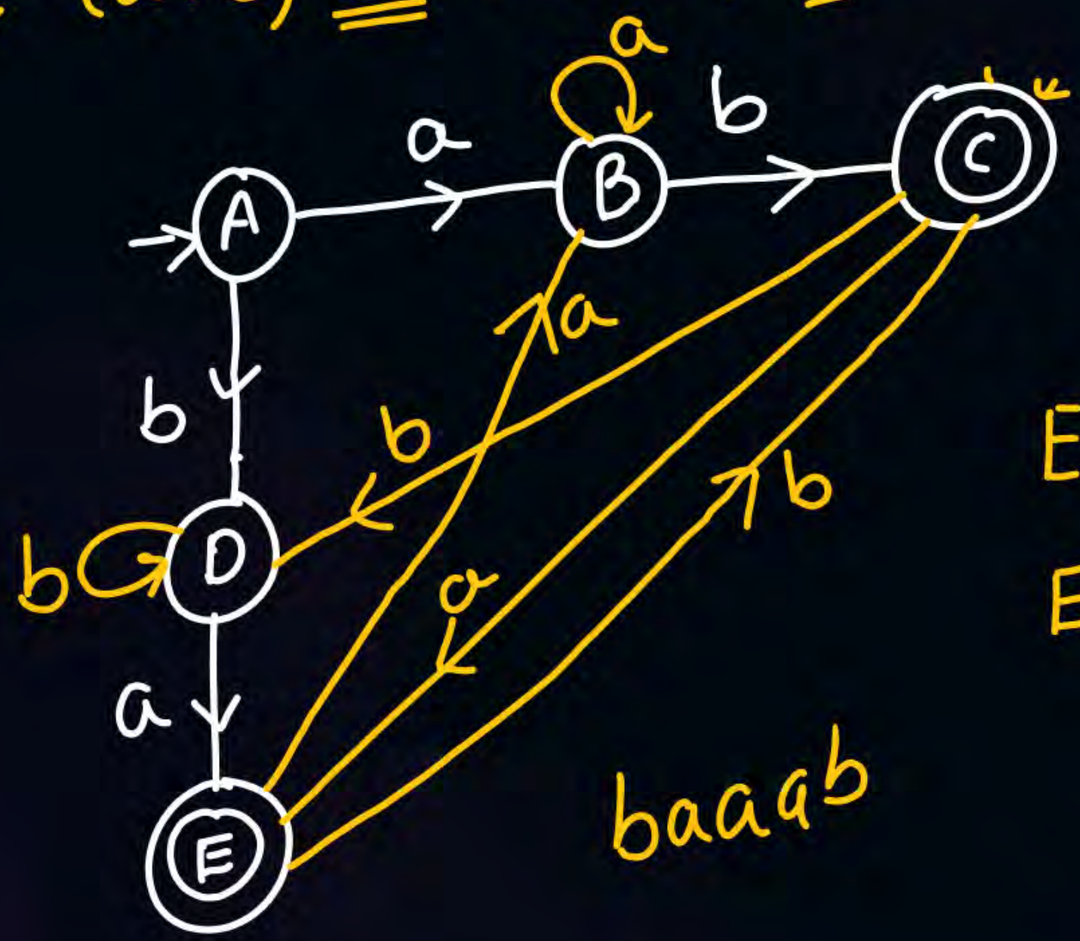
B → @b̸b

C → aa@b

E → bb a a

Last two symbols are different.

RE: $(a+b)^*ab + (a+b)^*ba$



A ✓
B ✓
D ✓

E → b ab
E → ba a b

baaab

First two symbols are diff

$ab(a+b)^* + ba(a+b)^*$



$\Sigma = \{0, 1\}$ $L = \{ w / w \text{ starts and ends with '11'} \}$

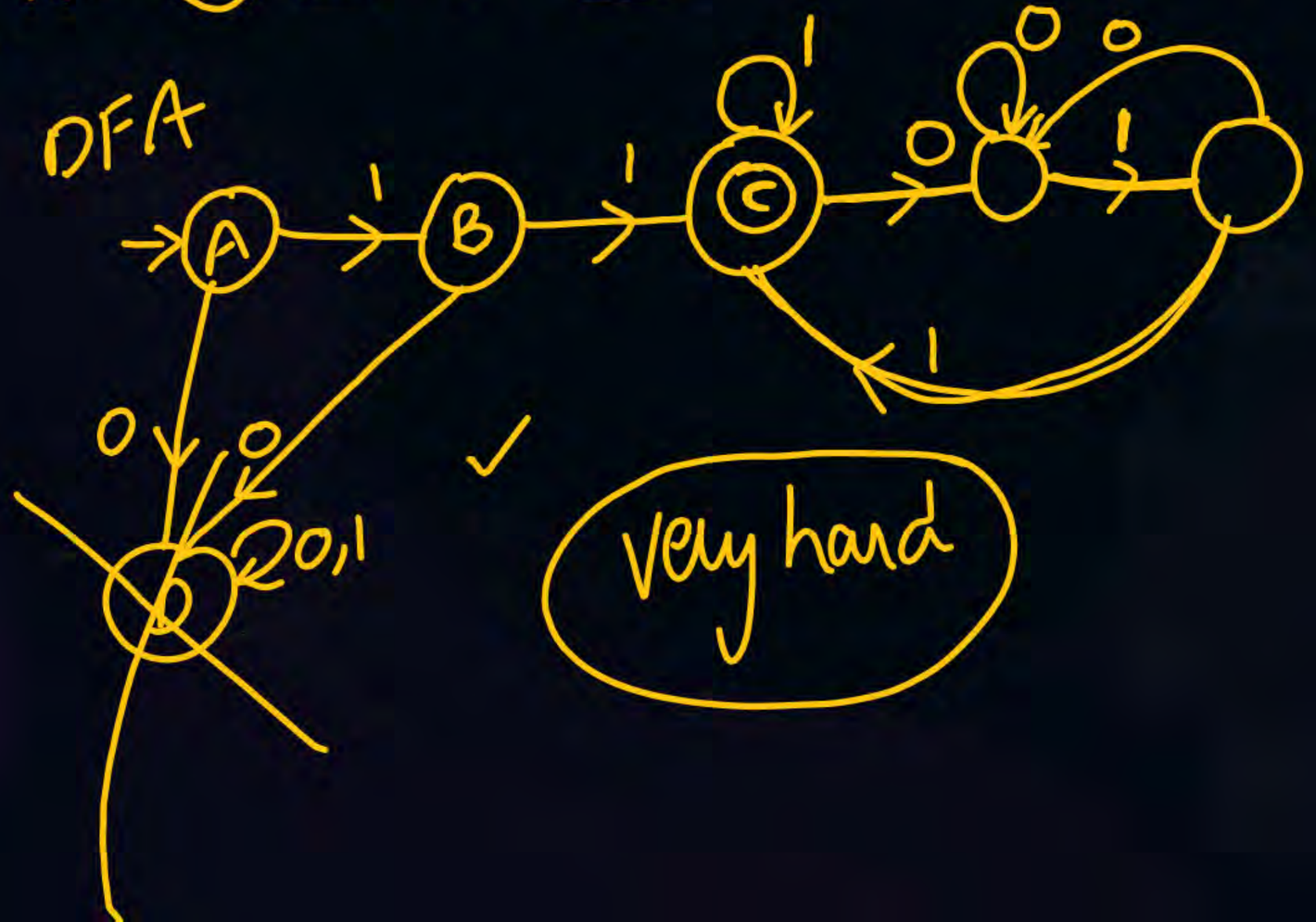
$L = \{ 11, 111, 1111, 11011, 11111 \ldots \}$

RE = $11 + 1111 + 11(0+1)^*11$

$11(0+1)^*11$

1111

DFA



very hard

$\Sigma = \{0, 1\}$ $L = \{ w / w$ starts with $\underline{10}$ and ends with $01 \}$

$101 + 10(1+0)^*01$

$L = \{ \underline{101}, 1001, 10101, 10001, 100001, \ldots\ldots \}$

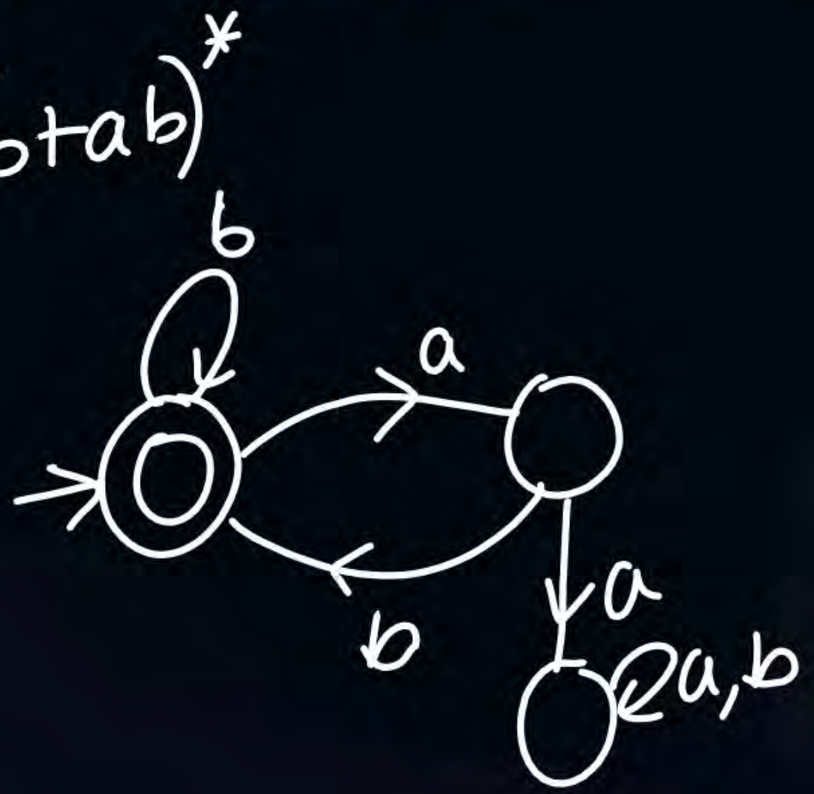


$\Sigma = \{a, b\}$

$L = \{w / \text{if } a \text{ is present in } w, \text{ then it always followed by a 'b'}\}$

$L = \{\epsilon, b, bb, bbb, \ldots\ldots, ab, bab, abab, abab, \ldots\ldots\}$

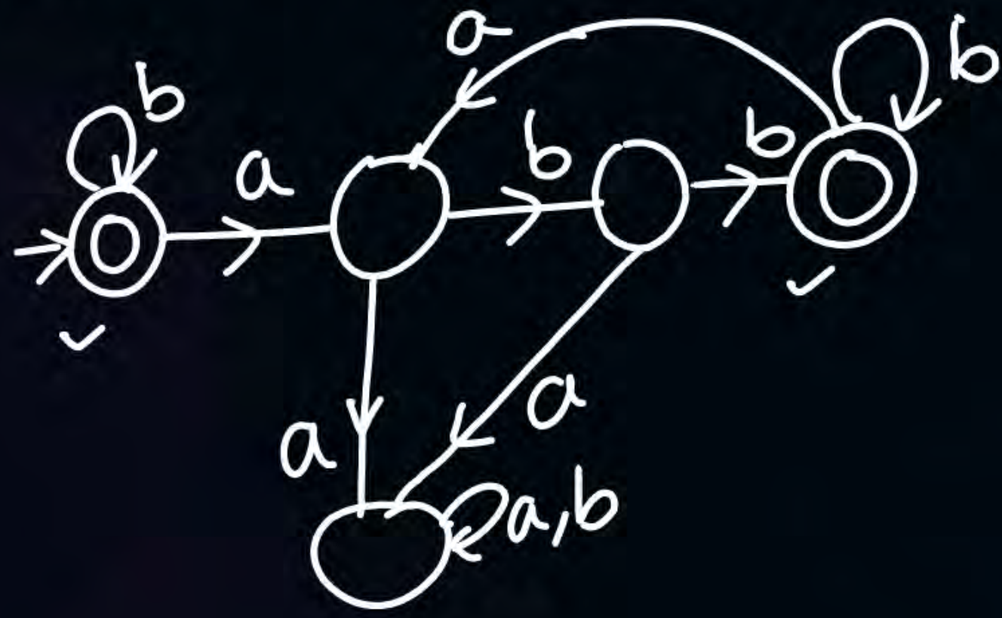
$(b + ab)^*$ $\quad$ $(b+ab)^*$

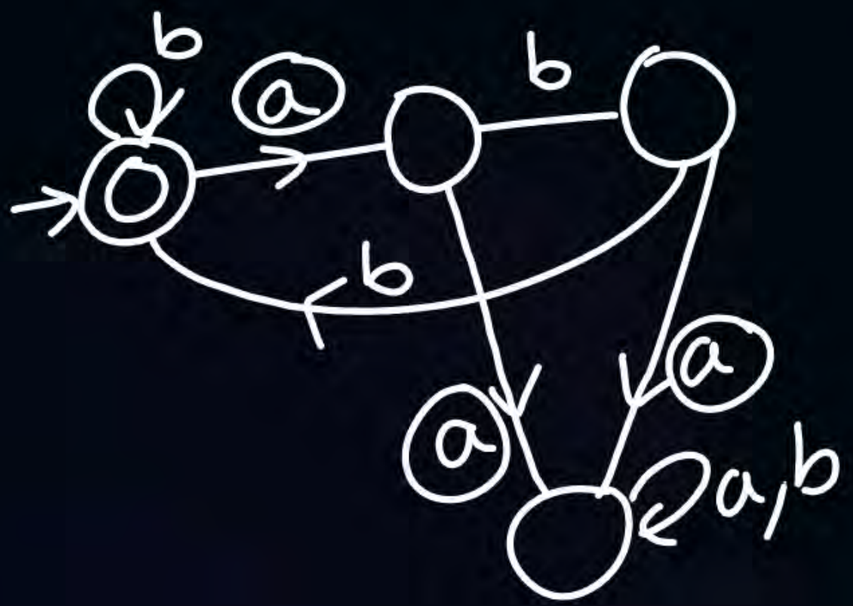


Every a is followed by 'bb'

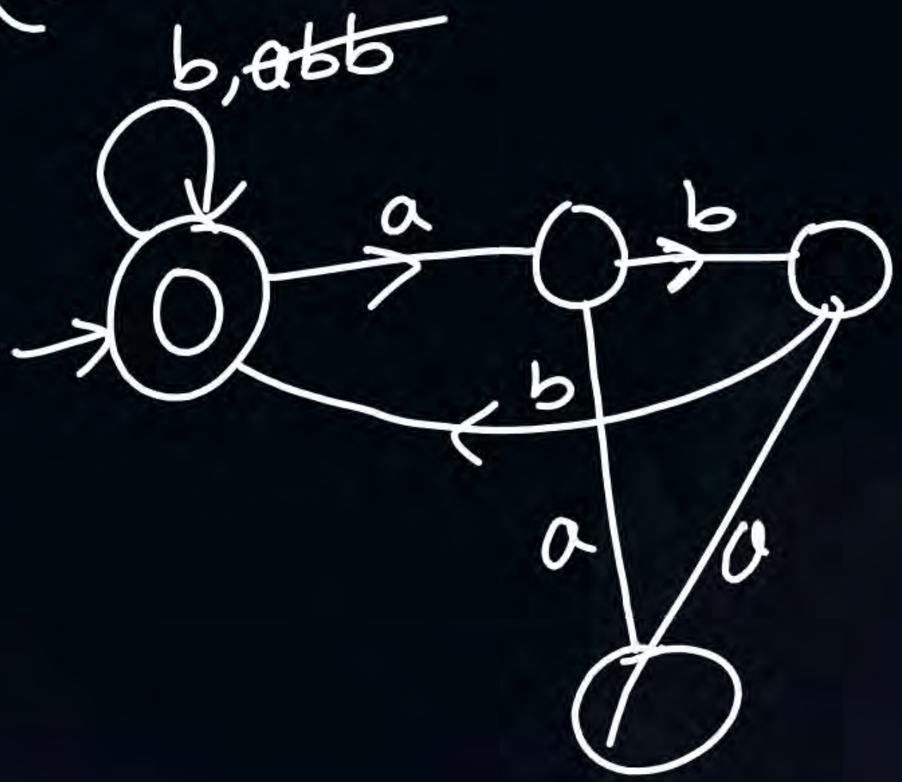
RE: $(b+abb)^*$

$L = \{\epsilon, b, bb, bbb, \cdots, abb, babb, abbb, \cdots\}$





~~a+~~ $(b+abb)^*$



$L = \{ w / $ if a is present in 'w', it should never be followed by 'b' $\}$

$L = \{ \underline{\epsilon, b, bb, bbb, \cdots} \; \underline{a, aa, aaa, \cdots}, ba, bba, baa \cdots \}$

$\underline{b^* a^*}$

$\underline{a^*}, \underline{b^*}, \underline{b^* a^*}$

$b^* a^*$

THANK - YOU